

E-Commerce Website features

1. User Authentication

Screens: Registration Form, Login Page, Forgot Password

Interaction:

- Secure sign-up and login with encryption
- Email verification on registration
- Password reset via email link
- CAPTCHA to block automated sign-ups

2. Profile & Address Management

Screens: Profile Overview, Edit Profile, Address Book

Interaction:

- View and update personal info and profile photo
- Change password securely
- Manage multiple delivery addresses

3. Role-Based Access Control

Screens: Role Selector, Admin Panel, Seller Dashboard

Interaction:

- Assign roles: Admin, Buyer, Optional Seller
- Role-based navigation and permissions
- Sellers can manage their own product listings

4. Product Catalog with Variants

Screens: Product Page, Variant Selector

Interaction:

- Products with size/color options
- Images and pricing change by variant
- Dynamic UI for variant selection

5. Advanced Search & Filtering

Screens: Search Results, Filter Panel

Interaction:

- Search with real-time suggestions
- Filter by category, price, brand, etc.
- Instant update of search results

6. Cart & Save for Later

Screens: Shopping Cart, Save for Later

Interaction:

- Add products with quantity controls
- Move items between cart and saved list
- Auto-update totals on cart changes

7. Wishlist Management

Screens: Wishlist, Product Page Shortcut

Interaction:

- Mark favorites for future purchase

- Move items from wishlist to cart
- Wishlist sale alerts via email/app

8. Ratings & Reviews

Screens: Product Reviews Section, Review Form

Interaction:

- Buyers submit ratings and comments
- Admin can moderate reviews
- Average rating shown on product

9. Flash Sales & Timed Deals

Screens: Flash Sale Banner, Countdown Widget

Interaction:

- Limited-time deals with countdown timer
- Timer resets post-expiry
- Boosts urgency and conversions

10. Promo & Discount Codes

Screens: Checkout Page, Admin Promo Panel

Interaction:

- Apply fixed or % promo codes
- Codes with expiry dates and usage limits
- Admin can manage and monitor codes

11. Multi-Payment Integration

Screens: Checkout, Payment Confirmation

Interaction:

- Pay via cards, bKash, Nagad, or COD
- Secure encryption at checkout
- Instant on-screen and email confirmation

12. Order Tracking

Screens: Order Status Tracker, Order Details

Interaction:

- Track: Confirmed → Shipped → Delivered
- Real-time status updates
- Shows estimated delivery window

13. Invoice & Billing

Screens: Invoice Page, Download Section

Interaction:

- Auto-generated PDF invoices
- View/download from user dashboard
- Includes full breakdown of orders

14. Admin Dashboard

Screens: Sales Overview, Product Manager, Reports Panel

Interaction:

- Manage users, orders, products, and coupons
- View stats and trends
- Export data for analysis

15. Notifications (Email & In-App)

Screens: Notifications Center, Email Templates

Interaction:

- In-app alerts for status updates
- Emails for order events and promotions
- Password change notifications

16. Smart Product Recommendations

Screens: Product Detail Page, Home Recommendations

Interaction:

- Suggest similar or related items
- Based on browsing/purchase behavior
- Enhances average order value

17. Customer Support & Contact

Screens: Contact Form, Admin Inbox

Interaction:

- Submit queries via support form
- Admin receives and replies via email
- Simple 4-field form (name, email, subject, message)

18. Chatbot Assistant

Screens: Chat Widget

Interaction:

- 24/7 rule-based assistant
- Answer FAQs, locate products, check order status
- Reduces load on support team

19. Loyalty & Reward System

Screens: Rewards Dashboard, Checkout Points Panel

Interaction:

- Earn points with each order
- Redeem points for discounts
- View balance and redemption history

20. Order History & Invoicing

Screens: Order History Page, Invoice Download

Interaction:

- Browse all previous orders
- Check status and download PDFs
- Includes price breakdown and shipping info

21. Returns & Refund Requests

Screens: Return Request Form, Request Status Page

Interaction:

- Submit return/refund request
- Admin reviews and responds
- Status updates visible to users

22. Gift Cards & Vouchers

Screens: Gift Card Purchase, Recipient Input

Interaction:

- Buy/send digital gift cards
- Recipient gets redeemable code
- Admin controls value and expiration

23. Activity Logs (Admin Only)

Screens: Admin Logs Viewer

Interaction:

- Track user/admin activity logs
- View login times, edits, purchases
- Supports issue resolution and auditing

24. Data Export (Admin Only)

Screens: Export Panel

Interaction:

- Export user/product/order data
- Choose CSV or PDF format
- Useful for reporting and backups

User Types

- **User**
- **Guest User**
- **Admin**
- **Delivery Man**

Short Description of user Types

Role	Description & Permissions
Guest User	Browse products, search/filter, view reviews, add to cart or wishlist (local), but cannot place orders.
User (Buyer)	Full user: login/register, manage profile, place orders, use wishlist/cart, earn rewards, track orders, write reviews, request returns/refunds.
Admin	Manage everything: users, orders, products, promotions, analytics, access logs, resolve return/refund, export data, reply to support.
Delivery Man	View assigned deliveries, update order status (e.g., “Out for delivery” → “Delivered”), see delivery address and contact.

Entity Overview

Type	Entity List
Core	User, Role, Profile, Address, AuthToken
Product	Product, Variant, Image, Category, Brand, Review
Order	Cart, Order, Invoice, Return, Refund, DeliveryStatus
Promotion	Coupon, GiftCard, FlashSale, PromoCodeUsage
Payment	Payment, Transaction, PaymentMethod
Support	SupportTicket, ChatMessage, Notification, ActivityLog
Recommendation	Recommendation, BrowsingHistory, SearchHistory
Rewards	RewardPoint, RewardTransaction
Optional	DeliveryPerson, DeliveryAssignment

Details of each entity

1. User

Column Name	Type	Description
id (PK)	UUID/INT	Unique user ID
name	VARCHAR	Full name
email	VARCHAR (unique)	Login email
phone	VARCHAR	Phone number
password_hash	TEXT	Encrypted password
is_verified	BOOLEAN	Email verification status
created_at	TIMESTAMP	Registration date

2. Role

Column Name	Type	Description
id (PK)	INT	Role ID
name	VARCHAR	Role name: Admin, Buyer, Seller

3. UserRole

Column Name	Type	Description
id (PK)	INT	ID
user_id (FK)	UUID/INT	References User(id)
role_id (FK)	INT	References Role(id)

4. Profile

Column Name	Type	Description
id (PK)	UUID/INT	Profile ID
user_id (FK)	UUID/INT	References User(id)
gender	VARCHAR	Gender
dob	DATE	Date of Birth
profile_photo	TEXT	Image URL or path
bio	TEXT	Short description

5. Address

Column Name	Type	Description
id (PK)	INT	Address ID
user_id (FK)	UUID/INT	References User(id)
recipient_name	VARCHAR	Name for delivery
phone	VARCHAR	Contact phone
address_line	TEXT	Street or address details
city	VARCHAR	City name
area	VARCHAR	Local area
postal_code	VARCHAR	ZIP or postal code
is_default	BOOLEAN	Default address flag

6. Product

Column Name	Type	Description
id (PK)	INT	Product ID
title	VARCHAR	Product name
description	TEXT	Detailed description
base_price	DECIMAL	Starting price
brand_id (FK)	INT	References Brand(id)
category_id (FK)	INT	References Category(id)
created_by (FK)	UUID/INT	References User(id) (seller)
created_at	TIMESTAMP	Date added

7. ProductVariant

Column Name	Type	Description
id (PK)	INT	Variant ID
product_id (FK)	INT	References Product(id)
variant_name	VARCHAR	e.g., Red - M, Blue - XL
price	DECIMAL	Variant price
stock	INT	Available quantity
sku	VARCHAR	Stock Keeping Unit

8. ProductImage

Column Name	Type	Description
id (PK)	INT	Image ID
product_id (FK)	INT	References Product(id)
variant_id (FK)	INT (nullable)	Optionally references ProductVariant(id)
image_url	TEXT	Image file path or URL

9. Category

Column Name	Type	Description
id (PK)	INT	Category ID
name	VARCHAR	Category name
parent_id (FK)	INT (nullable)	Self-reference for subcategory

10. Brand

Column Name	Type	Description
id (PK)	INT	Brand ID
name	VARCHAR	Brand name
logo_url	TEXT	Logo image
description	TEXT	Optional brand details

11. Review

Column Name	Type	Description
id (PK)	INT	Review ID
user_id (FK)	UUID/INT	Reviewer
product_id (FK)	INT	Reviewed product
rating	INT (1-5)	Star rating
comment	TEXT	Review text
created_at	TIMESTAMP	Review time
approved	BOOLEAN	Admin moderation status

12. Cart

Column Name	Type	Description
id (PK)	INT	Cart ID
user_id (FK)	UUID/INT	References User(id)

13. CartItem

Column Name	Type	Description
id (PK)	INT	Cart item ID
cart_id (FK)	INT	References Cart(id)
variant_id (FK)	INT	References ProductVariant(id)
quantity	INT	Selected quantity

14. Wishlist

Column Name	Type	Description
id (PK)	INT	Wishlist ID
user_id (FK)	UUID/INT	References User(id)
product_id (FK)	INT	References Product(id)
created_at	TIMESTAMP	Added date

15. Order

Column Name	Type	Description
id (PK)	INT	Order ID
user_id (FK)	UUID/INT	Buyer
address_id (FK)	INT	Delivery address
total_amount	DECIMAL	Total cost
status	VARCHAR	e.g., Confirmed, Shipped
created_at	TIMESTAMP	Order date

16. OrderItem

Column Name	Type	Description
id (PK)	INT	Order item ID
order_id (FK)	INT	References Order(id)
variant_id (FK)	INT	References ProductVariant(id)
quantity	INT	Quantity ordered
unit_price	DECIMAL	Price per unit

17. Invoice

Column Name	Type	Description
id (PK)	INT	Invoice ID
order_id (FK)	INT	References Order(id)
pdf_url	TEXT	Invoice file location
generated_at	TIMESTAMP	Date of generation

18. ReturnRequest

Column Name	Type	Description
id (PK)	INT	Return ID
order_item_id (FK)	INT	References OrderItem(id)
reason	TEXT	Return reason
status	VARCHAR	Pending/Approved/Rejected
created_at	TIMESTAMP	Submission time

19. Refund

Column Name	Type	Description
id (PK)	INT	Refund ID
return_id (FK)	INT	References ReturnRequest(id)
amount	DECIMAL	Refunded amount
method	VARCHAR	bKash/Card/etc.
status	VARCHAR	e.g., Processing, Done

20. Coupon

Column Name	Type	Description
id (PK)	INT	Coupon ID
code	VARCHAR	Unique code
type	VARCHAR	Flat / Percentage
value	DECIMAL	Discount value
expiry_date	DATE	Expiration date
usage_limit	INT	Max uses

21. GiftCard

Column Name	Type	Description
id (PK)	INT	Gift card ID
code	VARCHAR	Redeemable code
amount	DECIMAL	Gift card balance
recipient_email	VARCHAR	Receiver's email
sender_name	VARCHAR	Sender name
expiry_date	DATE	Expiration date

22. Payment

Column Name	Type	Description
id (PK)	INT	Payment ID
order_id (FK)	INT	References Order(id)
method	VARCHAR	bKash, Nagad, Card, COD
amount	DECIMAL	Amount paid
status	VARCHAR	Paid / Failed / Pending
transaction_id	VARCHAR	External gateway ID

23. Notification

Column Name	Type	Description
id (PK)	INT	Notification ID
user_id (FK)	UUID/INT	Recipient
type	VARCHAR	e.g., OrderUpdate, Promo
message	TEXT	Notification content
read	BOOLEAN	Read/unread
created_at	TIMESTAMP	Timestamp

24. AuthToken

Column Name	Type	Description
id (PK)	INT	Unique token ID
user_id (FK)	INT	References User(id)
token	TEXT	Authentication token
created_at	DATETIME	Token creation time
expires_at	DATETIME	Expiry time of the token

25. DeliveryStatus

Column Name	Type	Description
id (PK)	INT	Unique status ID
order_id (FK)	INT	References Order(id)
status	VARCHAR	e.g., Confirmed, Packed, Delivered
updated_at	DATETIME	Time of last status update
updated_by	VARCHAR	Role/person who updated the status

26. DeliveryPerson

Column Name	Type	Description
id (PK)	INT	Unique delivery person ID
name	VARCHAR	Full name
phone	VARCHAR	Contact number
email	VARCHAR	Email address
address	TEXT	Residential or base address
is_active	BOOLEAN	Availability status

27. DeliveryAssignment

Column Name	Type	Description
id (PK)	INT	Unique assignment ID
order_id (FK)	INT	References Order(id)
delivery_person_id (FK)	INT	References DeliveryPerson(id)
assigned_at	DATETIME	Assignment timestamp
status	VARCHAR	Assignment status (e.g., Assigned)

28. FlashSale

Column Name	Type	Description
id (PK)	INT	Unique flash sale ID
product_id (FK)	INT	References Product(id)
discount	DECIMAL	Discount value (flat or percentage)
start_time	DATETIME	Sale start time
end_time	DATETIME	Sale end time
is_active	BOOLEAN	Status of flash sale

29. PromoCodeUsage

Column Name	Type	Description
id (PK)	INT	Unique promo usage ID
user_id (FK)	INT	References User(id)
coupon_id (FK)	INT	References Coupon(id)
used_at	DATETIME	Timestamp when promo was used

30. Transaction

Column Name	Type	Description
id (PK)	INT	Unique transaction ID
payment_id (FK)	INT	References Payment(id)
request_data	TEXT	Raw data sent to gateway
response_data	TEXT	Response received
timestamp	DATETIME	Time of transaction

31. PaymentMethod

Column Name	Type	Description
id (PK)	INT	Unique method ID
name	VARCHAR	e.g., bKash, Card, COD
description	TEXT	Optional details
is_active	BOOLEAN	Whether method is available

32. SupportTicket

Column Name	Type	Description
id (PK)	INT	Ticket ID
user_id (FK)	INT	References User(id)
subject	VARCHAR	Ticket subject
message	TEXT	Initial user message
status	VARCHAR	e.g., Open, In Progress, Closed
created_at	DATETIME	Ticket creation time
resolved_at	DATETIME	When it was resolved (nullable)

33. ChatMessage

Column Name	Type	Description
id (PK)	INT	Message ID
user_id (FK)	INT	References User(id)
sender_type	VARCHAR	'user', 'admin', or 'bot'
message	TEXT	Chat content
timestamp	DATETIME	Message timestamp
support_ticket_id (FK)	INT	Optional, references SupportTicket(id)

34. ActivityLog

Column Name	Type	Description
id (PK)	INT	Log entry ID
user_id (FK)	INT	References User(id)
action	VARCHAR	e.g., Login, Logout, Edit Product
metadata	TEXT	Additional data (IP, device, etc.)
timestamp	DATETIME	When action happened

35. Recommendation

Column Name	Type	Description
id (PK)	INT	Unique recommendation ID
user_id (FK)	INT	References User(id)
product_id (FK)	INT	Suggested Product
reason	TEXT	e.g., "Based on browsing history"
score	DECIMAL	Confidence/priority score

36. BrowsingHistory

Column Name	Type	Description
id (PK)	INT	Unique record ID
user_id (FK)	INT	References User(id)
product_id (FK)	INT	Product viewed
viewed_at	DATETIME	Timestamp of view

37. SearchHistory

Column Name	Type	Description
id (PK)	INT	Unique record ID
user_id (FK)	INT	References User(id)
query	VARCHAR	Search query entered
searched_at	DATETIME	When search was made

38. RewardPoint

Column Name	Type	Description
id (PK)	INT	Reward record ID
user_id (FK)	INT	References User(id)
total_points	INT	Total available points
last_updated	DATETIME	Last update timestamp

39. RewardTransaction

Column Name	Type	Description
id (PK)	INT	Transaction ID
user_id (FK)	INT	References User(id)
order_id (FK)	INT	References Order(id), if applicable
points	INT	Earned (positive) or spent (negative)
reason	VARCHAR	e.g., "Purchase", "Redemption"
timestamp	DATETIME	Time of transaction

Workflow

0. Initial Setup & Roles

Before coding anything:

- **Decide roles** (everyone can help each other too, but it helps to assign ownership):
 - **Lead Backend Dev (Entity & DB design)** → e.g., Person A
 - **API Developer (Routes, Controllers)** → e.g., Person B
 - **Frontend Dev (UI/UX, API integration)** → e.g., Person C
 - **Project Manager / QA / Docs / Integration Support** → e.g., Person D
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1. Version Control Setup

- Use **Git + GitHub** (or GitLab, Bitbucket) — create a shared repo.
 - Setup main branches:
 - **main** – production-ready
 - **dev** – all features merged here
 - **feature/xyz** – each task/feature gets its own branch
 - Set up **code reviews via pull requests**.
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2. Project Structure (Backend)

Choose stack first (e.g., **Node.js (Express) + Sequelize, Django, Spring Boot**, etc.)

Let's assume **Node.js + Express + Sequelize** as an example.

Suggested folder structure:

bash

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/backend

```
|— /models      ← All entity classes (based on your DB schema)
|— /routes      ← Route definitions (e.g., /user, /product)
|— /controllers ← Business logic per route
|— /middlewares ← Auth, error handling, etc.
|— /utils       ← Helpers (email, token, etc.)
|— /config      ← DB config, env vars
|— index.js     ← App entry point
```

3. Entity Classes First (Yes!)

Yes — **start by creating all Entity Classes (Models)** using Sequelize (or any ORM of your stack).

One member (or two working in parallel) can:

- Create models for all 39 tables you designed.
 - Setup proper associations: `belongsTo`, `hasMany`, `hasOne`.
 - Create migrations if using Sequelize CLI or similar.
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4. Database Initialization

Once models are ready:

- Write code to sync with DB (`sequelize.sync()` or migrations).
 - Use **seeders** to populate default roles, categories, brands, etc.
 - Test it locally to make sure the DB is created as expected.
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5. API Design & Routes

While models are being worked on, another team member can start designing **API endpoints**:

Example endpoints:

- `/auth/register`, `/auth/login`
- `/product`, `/product/:id`
- `/cart`, `/wishlist`

- /admin/approve-review, etc.

You can use a tool like **Postman** to test.

Define routes in:

- routes/userRoutes.js
- routes/productRoutes.js, etc.

🧠 6. Frontend Developer Starts With UI Mockups

The frontend person can start building pages like:

- Registration/Login
- Product Listing
- Cart/Wishlist
- Order Summary
- Dashboard (Admin/Seller)

Mock APIs can be used at first; later they will integrate with real backend.

📅 7. Weekly Planning (Class Time)

Since you meet in class:

- Use class time to:
 - Assign tasks for the week
 - Demo your progress
 - Fix bugs together
 - Clarify logic and discuss blockers

Use tools like:

- **Trello / Notion / GitHub Projects** → for task management
- **WhatsApp/Discord/Slack** → daily quick updates

☑ 8. Testing + QA

- One member can take responsibility to test flows:
 - Register → Login → Add to Cart → Order → Payment
 - Admin Actions → Review Management → Delivery Assignment
- Use tools like Postman for API testing, or browser dev tools for frontend.

📄 9. Docs & Final Report

Start documentation early:

- ER diagram
- Workflow diagrams
- API docs (Swagger or manually written)
- Screenshots of UI
- Tech stack used

Summary Workflow

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Week 1:

- ✓ Git + Project Structure
- ✓ Entity/Model Classes
- ✓ DB Setup

Week 2:

- ✓ Auth API + Basic Routes
- ✓ Frontend starts UI

✓ Seeders + Roles

Week 3:

✓ Cart/Wishlist/Order APIs

✓ Frontend → API Integration

✓ Flash Sale, Payment, Notification

Week 4+:

✓ Delivery + Chat + Admin Dashboard

✓ Testing, Polish UI, Docs

✓ Final Integration & Presentation